SPURVANCE

Internship Program

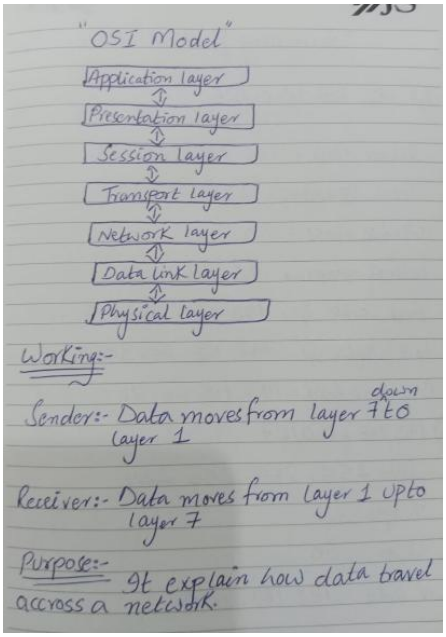
Assignment

Submitted By: Maryam Rehan

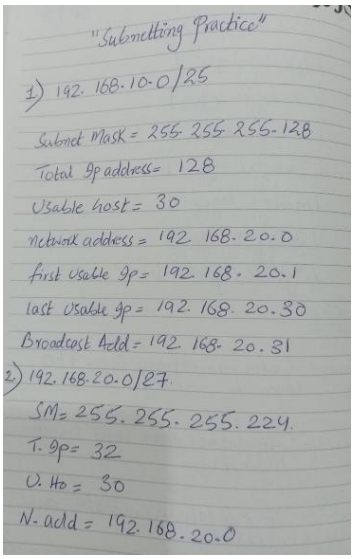
Mentor: Miss Aroosa Tariq

OS & NetworkingFundamentals

Draw the OSI model with examples:



Subnetting practice sheet:

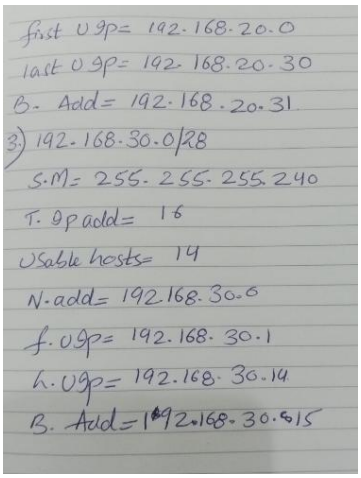

"Subnetting Practice"

1) 192.168.10.0/25

Subnet Mask = 255.255.255.128
Total Ip address= 128
Usable host = 30
network address = 192.168.20.0
first usable Ip = 192.168.20.1
last usable Ip = 192.168.20.30
Broadcast Add = 192.168.20.31

2) 192.168.20.0/27

SM = 255.255.255.224
T. Ip = 32
U. Ho = 30
N. add = 192.168.20.0


first U Ip = 192.168.20.0
last U Ip = 192.168.20.30
B. Add = 192.168.20.31

3) 192.168.30.0/28

S.M = 255.255.255.240
T. Ip add = 16
Usable hosts = 14
N. add = 192.168.30.0
f. Uip = 192.168.30.1
L. Uip = 192.168.30.14
B. Add = 192.168.30.15

Demo of 10 Linux commands:

```
┌──(root㉿kali)-[~]
└─# pwd
/root
```

Displays the current working directory

```
┌──(root㉿kali)-[~]
└─# ls
cookies.txt  Downloads  Mediroza  Module1
```

Current directory files folders Listing

```
┌──(root㉿kali)-[~]
└─# whoami
root
```

Current logged-in username show

```
┌──(root㉿kali)-[~]
└─# date
Thu Oct 15 12:54:15 PM EDT 2026
```

System date time show

```
┌──(root㉿kali)-[~]
└─# mkdir demo
```

Demo folder create

```
┌──(root㉿kali)-[~]
└─# cd demo

┌──(root㉿kali)-[~/demo]
└─#
```

Inside Demo folder

```
┌──(root㉿kali)-[~/demo]
└─# touch file.txt
```

Empty text file create

```
┌──(root㉿kali)-[~/demo]
└─# echo "Hello Linux" > file.txt
```

Write text inside file

```
┌──(root㉿kali)-[~/demo]
└─# cat file.txt
Hello Linux
```

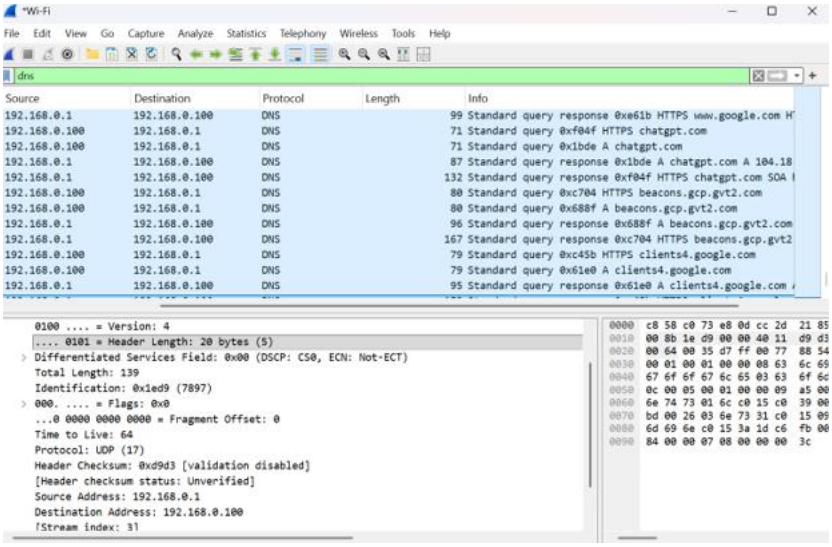
Display Text inside file

```
┌──(root㉿kali)-[~/demo]
└─# cp file.txt copy.txt
```

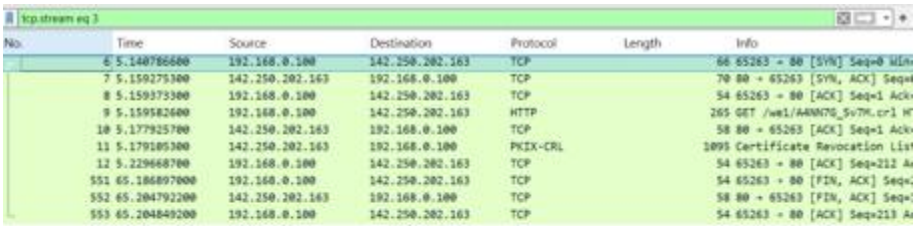
Create Copy of file

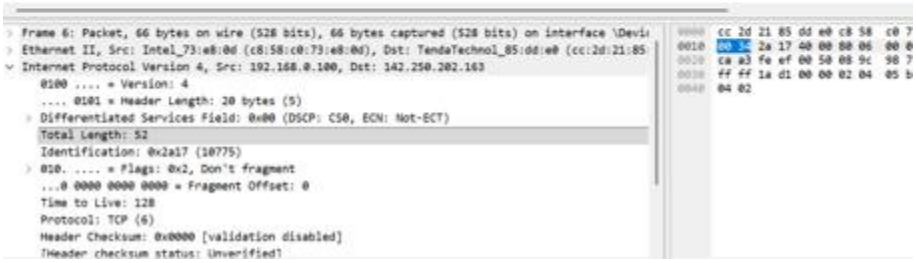
<u>WIRESHARK:</u>

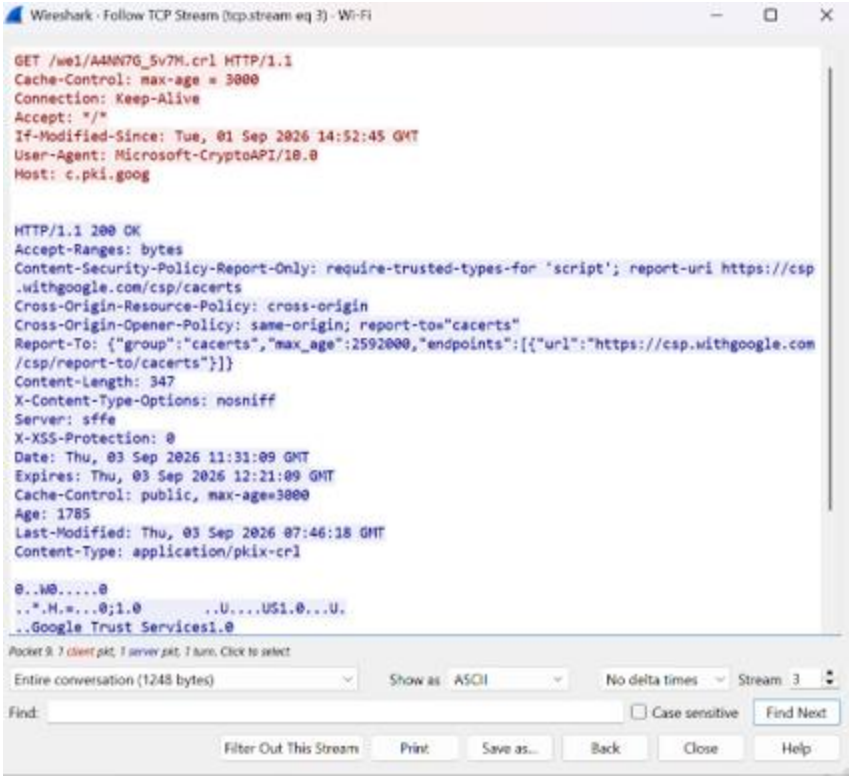
DNS PACKET


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
Source Destination Protocol Length Info
192.168.0.1 192.168.0.100 DNS 99 Standard query response 0xe61b HTTPS www.google.com H
192.168.0.100 192.168.0.1 DNS 71 Standard query 0xf04f HTTPS chatgpt.com
192.168.0.100 192.168.0.1 DNS 71 Standard query 0x1bde A chatgpt.com
192.168.0.1 192.168.0.100 DNS 87 Standard query response 0x1bde A chatgpt.com A 104.18
192.168.0.1 192.168.0.100 DNS 132 Standard query response 0xf04f HTTPS chatgpt.com SOA
192.168.0.100 192.168.0.1 DNS 80 Standard query 0xc704 HTTPS beacons.gcp.gvt2.com
192.168.0.100 192.168.0.1 DNS 80 Standard query 0x688f A beacons.gcp.gvt2.com
192.168.0.1 192.168.0.100 DNS 96 Standard query response 0x688f A beacons.gcp.gvt2.com
192.168.0.1 192.168.0.100 DNS 167 Standard query response 0xc704 HTTPS beacons.gcp.gvt2
192.168.0.100 192.168.0.1 DNS 79 Standard query 0xc45b HTTPS clients4.google.com
192.168.0.100 192.168.0.1 DNS 79 Standard query 0x61e0 A clients4.google.com
192.168.0.1 192.168.0.100 DNS 95 Standard query response 0x61e0 A clients4.google.com
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 139
Identification: 0x1ed9 (7897)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0xd9d3 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.0.1
Destination Address: 192.168.0.100
[Stream index: 3]
0000 c8 58 c0 73 e8 0d cc 2d 21 85
0010 00 8b 1e d9 00 00 40 11 d9 d3
0020 00 64 00 35 d7 ff 00 77 88 54
0030 00 01 00 01 00 00 08 63 6c 69
0040 67 6f 6f 67 6c 65 03 63 6f 6d
0050 0c 00 05 00 01 00 00 09 a5 00
0060 6e 74 73 01 6c c0 15 c0 39 00
0070 bd 00 26 03 6e 73 31 c0 15 09
0080 6d 69 6e c0 15 3a 1d c6 fb 00
0090 84 00 00 07 08 00 00 00 3c

HTTP PACKET


tcp.stream eq 3
No. Time Source Destination Protocol Length Info
6 5.140786600 192.168.0.100 142.250.202.163 TCP 66 65263 → 80 [SYN] Seq=0 Win
7 5.159275300 142.250.202.163 192.168.0.100 TCP 70 80 → 65263 [SYN, ACK] Seq=
8 5.159373300 192.168.0.100 142.250.202.163 TCP 54 65263 → 80 [ACK] Seq=1 Ack=
9 5.159582600 192.168.0.100 142.250.202.163 HTTP 265 GET /we1/A4NN7G_5vTM.crl HT
10 5.177925700 142.250.202.163 192.168.0.100 TCP 58 80 → 65263 [ACK] Seq=1 Ack=
11 5.179105300 142.250.202.163 192.168.0.100 PKIX-CRL 1095 Certificate Revocation List
12 5.229668700 192.168.0.100 142.250.202.163 TCP 54 65263 → 80 [ACK] Seq=212 A
551 65.186897000 192.168.0.100 142.250.202.163 TCP 54 65263 → 80 [FIN, ACK] Seq=
552 65.204792200 142.250.202.163 192.168.0.100 TCP 58 80 → 65263 [FIN, ACK] Seq=
553 65.204849200 192.168.0.100 142.250.202.163 TCP 54 65263 → 80 [ACK] Seq=213 A


Frame 6: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Devi
Ethernet II, Src: Intel_73:e8:0d (c8:58:c0:73:e8:0d), Dst: TendaTechnol_85:dd:e0 (cc:2d:21:85
Internet Protocol Version 4, Src: 192.168.0.100, Dst: 142.250.202.163
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 52
Identification: 0x2a17 (10775)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
0000 cc 2d 21 85 dd e0 c8 58 c0 7
0010 00 34 2a 17 40 00 80 06 00 0
0020 ca a3 fe ef 00 50 08 9c 98 7
0030 ff ff 1a d1 00 00 02 04 05 b
0040 04 02



Source IP: 192.168.0.100
Destination IP: 142.250.202.163
Protocol: HTTP
Source Port: 65263
Destination Port: 80
Request: GET /wel/A4NN7G_5v7M.crl
Packet Length: 265 bytes

Analysis: The capture shows an HTTP GET request from the client to a remote server over TCP port 80. The TCP three-way handshake is visible before the HTTP request. The server then sends a Certificate Revocation List (CRL).

Purpose: Wireshark helps identify HTTP requests and analyze communication between a client and a web server.